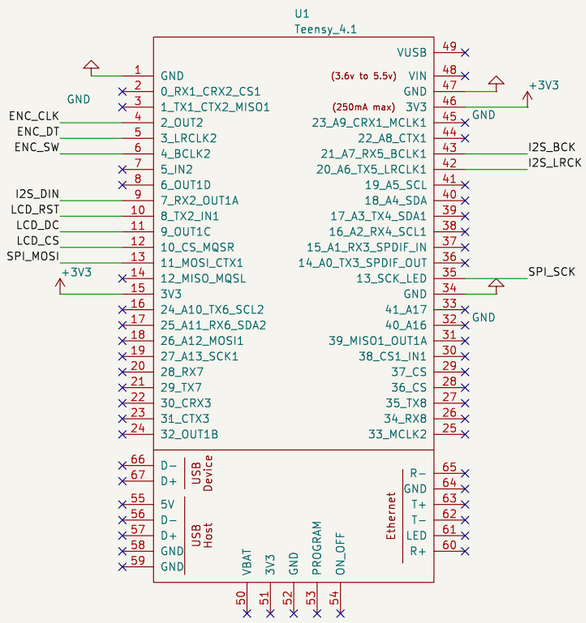
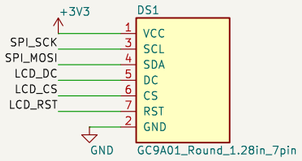
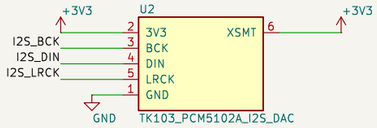








Teensy 4.1 front panel on an ElectroCookie full-size breadboard PCB.
Pin assignments are the ones tested in cheap-comp-midi-pcb (ui/README.md, xtragravy/bench/teensy41/README.md).
Teensy socket: left header in C1-C24, right header in G1-G24, micro-USB at row 1, micro-SD at row 24.
Teensy pin -> ElectroCookie hole (use the free holes of the same row strip):
  GPIO 2/3/4   (encoder CLK/DT/SW) -> A4/A5/A6   GPIO 7 (I2S DIN) -> A9
  GPIO 8/9/10 (LCD RST/DC/CS)   -> A10/A11/A12   GPIO 11 (MOSI)   -> A13   GPIO 13 (SCK) -> J14
  GPIO 20/21   (I2S LRCK/BCK)   -> J7/J6   3.3 V -> A15 or J3   GND -> A1, J2 or J15
TK103 XSMT is pulled high on the module; tied to 3.3 V here as in the tested build (a GPIO could mute it).
All modules run from the Teensy 3.3 V regulator (250 mA max): display backlight + DAC + encoder pull-ups are well under that.

Generated by tools/gen_teensy_panel.py; pins per cheap-comp-midi-pcb Teensy build

Sheet: /
File: teensy-panel.kicad_sch

Title: Teensy 4.1 front panel: GC9A01 + KY-040 + TK103 I2S DAC

Size: A3	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 1/1